

Machine Learning - Mid-Semester 1

Mock 01 Solutions

Contents

Machine Learning Midsem 1 - Mock 01 Answer Key	3
Q1 (4 marks)	3
Q2 (4 marks)	3
Q3 (4 marks)	3
Q4 (4 marks)	3
Q5 (4 marks)	3
Q6 (4 marks)	4
Q7 (3 marks)	4
Q8 (3 marks)	4
Machine Learning Midsem 1 - Mock 01 Rubric	4
Blueprint audit	8

Machine Learning Midsem 1 - Mock 01 Answer Key

Q1 (4 marks)

- 1 T: binary prediction of pump failure within seven days. E: historical sensor sequences paired with whether failure occurred in the following seven days. P: a declared held-out measure such as recall, precision-recall trade-off, or expected error cost. The measure must match operational harm. (2)
- 2 This is supervised binary classification because examples have outcome labels. If historical pumps, operating loads, or failure labels do not represent future deployment, held-out performance on that same biased source does not justify future performance. (2)

Q2 (4 marks)

- 1 Precision uses predicted positives: $21/(21+9)=21/30=0.70$. Recall uses actual positives: $21/(21+7)=21/28=0.75$. (2)
- 2 Scaling on all records leaks validation/test distribution into fitting. First form valid train, validation, and test splits. Fit scaler and model only on training; apply the frozen scaler to validation for choices; freeze pipeline and threshold; apply both once to test. (2)

Q3 (4 marks)

- 1 The normal-equation system is $X^T X w = X^T y$; it can be solved without explicitly forming an inverse. The direct-inverse expression is $w = (X^T X)^{-1} X^T y$, and it requires full column rank plus acceptable conditioning. Otherwise use a stable least-squares factorization or pseudoinverse. (1)
- 2 X is 3 by 2, w is 2 by 1, and y is 3 by 1.

$$X^T X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \quad X^T y = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

shapes: 2 by 2 and 2 by 1

The equations are $3w_0+3w_1=9$ and $3w_0+5w_1=13$. Subtracting gives $2w_1=4$, so $w_1=2$; then $w_0=1$. Thus $w=(1,2)^T$, predicting $(1,3,5)^T$. (3)

Q4 (4 marks)

- 1 The boundary is the set of x satisfying $w^T x + b = 0$. Score is $3(2)+4(1)-10=0$, so the point lies on the boundary. Since $\|w\|_2 = \sqrt{3^2+4^2}=5$, signed distance is $0/5=0$. (2)
- 2 $P(\text{disease}|x)=0.15$ and $P(\text{healthy}|x)=0.85$. Risk(treat) is $2(0.85)=1.70$; risk(do not treat) is $20(0.15)=3.00$, so treat. The score ranks/splits, normalized score gives geometric distance, posterior describes belief, and the action minimizes cost; none may be silently substituted for another. (2)

Q5 (4 marks)

- 1 Binary $p = \sigma(z) = 1/(1 + \exp(-z))$ and $\ln[p/(1-p)] = z = w^T x + b$. Convexity applies to cross-entropy for a linear logistic specification with a convex L1/L2 penalty, not to an arbitrary neural network; uniqueness can still depend on rank and regularization. (2)
- 2 Exponentials are (2,1,5), sum 8, so probabilities are (1/4,1/8,5/8). The true class is first, so loss is $-\ln(1/4) = \ln 4$, approximately 1.3863. (2)

Q6 (4 marks)

- 1 $H = -\sum_k p_k \log_2(p_k)$, with $0 \log 0 = 0$. Parent proportions (1/2,1/2) give $H=1$. Each child has proportions (4/5,1/5) in opposite order:

$$H_{\text{child}} = -(4/5)\log_2(4/5) - (1/5)\log_2(1/5) = 0.7219 \text{ bits.}$$

Equal sizes give weighted child entropy 0.7219, so gain is $1 - 0.7219 = 0.2781$ bit. (3)

- 1 Valid policies include learned left/right routing, surrogate splits, probabilistic routing, or training-fitted imputation. A tree can sort continuous values and evaluate threshold tests $x_j \leq t$; prior binning is not required. (1)

Q7 (3 marks)

- 1 Uniform labels A,B,B give B. Inverse-distance totals are $A = 1/1 = 1$ and $B = 1/2 + 1/3 = 5/6$, so weighted voting gives A. Distances are positive, so division is defined. (1)
- 2 kNN directly aggregates selected stored targets. LWR fits a weighted query-specific regression. An RBF model maps distances from centres into features and fits global coefficients rather than answering by neighbour vote. (2)

Q8 (3 marks)

- 1 The team's newest month is not an honest test because repeated feature, hyperparameter, or threshold choices adapt to it. Instead, sort by time: use an earlier period for training, a later period for validation choices, and the newest untouched period for final test. Fit imputation/scaling/features/model on training; choose hyperparameters and threshold on validation; freeze everything; evaluate once on test. If a purported test month has already guided choices, treat it as validation data and obtain a new future test period. (2)
- 2 A threshold implements the chosen cost/metric and is therefore a validation choice; tuning it on test consumes the test. Recall emphasizes the fraction of actual fraud detected; an explicit expected-cost metric is even closer when costs are known. (1)

Machine Learning Midsem 1 - Mock 01 Rubric

Score each point independently. Follow-through credit is allowed when a later operation correctly uses an explicitly stated earlier arithmetic result, but no credit is awarded for an invalid method or missing required condition.

Point	Marks	Primary session	Form	Difficulty	Objective criterion
Q1.1a	1	1	definition	Easy	Correct task and experience, including seven-day labeled outcome.
Q1.1b	1	1	definition	Easy	Performance measure is operationally meaningful and correctly described.
Q1.2a	1	1	scenario	Standard	Identifies supervised binary classification with reason.
Q1.2b	1	1	scenario	Standard	Explains representative-experience failure without causal overclaim.
Q2.1a	1	2	numerical	Easy	Precision 0.70 with correct denominator.
Q2.1b	1	2	numerical	Standard	Recall 0.75 with correct denominator.
Q2.2a	1	2	scenario	Standard	Identifies all-data scaling as leakage.
Q2.2b	1	2	scenario	Stretch	Gives complete fit-train, choose-validation, final-test order.

Point	Marks	Primary session	Form	Difficulty	Objective criterion
Q3.1	1	3	comparison	Stretch	Distinguishes the equation system from explicit inversion and gives the rank /conditioning condition.
Q3.2a	1	3	numerical	Easy	Correct shapes and $X^T X$.
Q3.2b	1	3	numerical	Standard	Correct $X^T y$ and simultaneous equations.
Q3.2c	1	3	numerical	Standard	Solves $w=(1,2)^T$ and verifies prediction.
Q4.1a	1	4	definition	Easy	Defines $g(x)=0$ boundary and obtains score zero.
Q4.1b	1	4	numerical	Standard	Uses 'g/
Q4.2a	1	4	numerical	Standard	Risks 1.70 and 3.00; chooses treat.
Q4.2b	1	4	scenario	Stretch	Correctly separates score, distance, posterior, and action.
Q5.1a	1	5	definition	Easy	Correct sigmoid and log-odds relationship.

Point	Marks	Primary session	Form	Difficulty	Objective criterion
Q5.1b	1	5	comparison	Stretch	Distinguishes convex linear logistic loss with convex penalty from an arbitrary neural-network objective.
Q5.2a	1	5	numerical	Standard	Softmax probabilities (1/4, 1/8, 5/8).
Q5.2b	1	5	numerical	Standard	Cross-entropy $\ln 4$ or accurate equivalent.
Q6.1a	1	6	definition	Easy	Defines entropy with $0 \log 0 = 0$; parent entropy 1.
Q6.1b	1	6	numerical	Standard	Each child entropy approximately 0.7219.
Q6.1c	1	6	numerical	Standard	Weighted gain approximately 0.2781 bit.
Q6.2	1	6	scenario	Stretch	Valid missing policy and threshold handling for continuous feature.
Q7.1	1	7	numerical	Easy	Uniform predicts B; inverse-distance totals 1 and 5/6 predict A.

Point	Marks	Primary session	Form	Difficulty	Objective criterion
Q7.2a	1	7	scenario	Standard	Distinguishes direct kNN aggregation from query-local LWR fitting.
Q7.2b	1	7	scenario	Stretch	Distinguishes a global coefficient-fitted RBF model from neighbour voting and local fitting.
Q8.1a	1	8	comparison	Easy	Contrasts repeated newest-month tuning with a chronological split and training-only transformation fit.
Q8.1b	1	8	comparison	Standard	Distinguishes validation choice-making from a fresh one-time final test.
Q8.2	1	8	scenario	Standard	Places threshold tuning before test and justifies recall/cost metric.

Blueprint audit

- Sessions 1–8: 4,4,4,4,4,3,3 marks = 30.
- Forms: property/definition 5; comparison/distinction 4; numerical reasoning 12; scenario/interpretation 9 = 30.
- Difficulty: Easy 9; Standard 15; Stretch 6 = 30 (30%/50%/20%, exact largest-remainder allocation).